WW-P-471B
May 21, 1970
SUPERSEDING
Fed. Spec. WW-P-471A
September 27, 1960

FEDERAL SPECIFICATION

PIPE-FITTINGS: BUSHINGS, LOCKNUTS, AND PLUGS; IRON, STEEL, AND ALUMINUM; (THREADED); 125-150 POUNDS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for bushings, locknuts, and plugs made of aluminum, iron, or steel.

1.2 Classification.

1.2.1 Types and styles. Bushings, locknuts, and plugs shall be of the following types, kind, and styles, as specified (see 6.2):

Type II - Iron or steel.
- Style A - Uncoated.
- Style C - Zinc coated.
- Style D - Cadmium plated.

Type III - Aluminum
- Style A - Anodized.
- Style B - Nonanodized.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

FSC 4730

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Federal Specifications:

QQ-A-225	-	Aluminum Alloy Bar, Rod, Wire or Special Shapes, Rolled Drawn, or Cold Finished, General Specification for.
QQ-A-225/1c	-	Aluminum Alloy Bar, Rod, and Wire, Rolled Drawn, or Cold Finished, 1100.
QQ-A-225/3c	-	Aluminum Alloy Bar, Rod, and Wire, 2011.
QQ-A-225/6c	-	Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn or Cold Finished, 2024.
QQ-I-652	-	Iron Castings, Gray.
QQ-I-666	-	Iron Castings, Malleable.
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
QQ-S-630	-	Steel Bar, Carbon, Hot Rolled.
QQ-S-681	-	Steel Castings.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.

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MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI).

B2.1 - Pipe Threads (Except Dryseal).
B16.14 - Ferrous Pipe Plugs, Bushings, and Locknuts with Pipe Threads.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

American Society for Testing and Materials (ASTM) Standards:

A126 - Castings for Valves, Flanges and Pipe Fittings.
A153 - Zinc-Coating (Hot-Dip) on Iron and Steel Hardware.
A165 - Electrodeposited Coatings of Cadmium on Steel.
A197 - Malleable Iron Castings.
A216 - Carbon Steel Castings Suitable for Fusion Welding.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Bureau of Standards (NBS) Handbook:

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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Society of Automotive Engineers (SAE).

J531 - Automotive Pipe, Filler and Drain Plugs.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N.Y. 10017.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Atten: Tariff Order Section. 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, I11. 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material.

3.1.1 Iron and steel. Type II steel fittings shall conform to QQ-S-681, class 65-35 or 70-36; or to QQ-S-630; or ASTM A216. Type II iron fittings shall conform to QQ-I-652, class 20 to class 30 inclusive to ASTM A126; or to QQ-I-666, grades I or II; or ASTM A197.

3.1.2 Aluminum. Type III (aluminum) fittings shall conform to QQ-A-225/1c, QQ-A-225/3c or QQ-A-225/6c.

3.2 Standard product. Fittings shall be the standard commercial product of the manufacturer. The items furnished shall be new and unused.

3.3 Design. Fittings shall be furnished in the types, kinds, and sizes specified (see 6.2), and shall be in accordance with ANSI B16.14, and SAE J531.

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3.4 Dimensions. Dimensions for 1 inch and smaller size plugs shall be in accordance with table 4 of SAE J531 or ANSI B16.14. Plugs in sizes 1¼ inch and larger shall be in accordance with ANSI B16.14.

3.5 Finish.

3.5.1 Zinc-coated. When specified (see 6.2), iron and steel fittings shall be zinc-coated in accordance with ASTM A153. The zinc-coating shall be applied prior to all threading.

3.5.2 Cadmium-plated. When specified (see 6.2), the iron and steel fittings shall be cadmium-plated in accordance with QQ-P-416 or ASTM A165.

3.5.3 Anodized. When specified (see 1.2.1 and 6.2), aluminum fittings shall be anodized in accordance with MIL-A-8625.

3.6 Threads. Unless otherwise specified (see 6.2), all threads shall be right-hand and shall be smooth, clean, and true to form. Threads shall be in accordance with NBS H28 or ANSI B2.1.

3.6.1 Bushings. When specified (see 6.2 and 6.3), outside head bushings shall be externally threaded and gaged slightly oversized.

3.6.2 Locknuts. All locknuts shall have internal threads in accordance with NBS H28 or ANSI B2.1.

3.7 Identification marking. The name or initial of the manufacturer, or other mark by which he can be readily identified, shall be legibly cast or die-stamped on each fitting, unless the size or shape does not permit such marking to be applied.

3.8 Workmanship. Fittings shall be thoroughly cleaned, made free of sand and scale, be sound, smoothly cored, true to form, and free of cold shuts, blowholes, porosity, or any other defect that may affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier

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may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested in accordance with the requirements of referenced subsidiary specifications and standards, if any.

4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except when otherwise specified herein. For purposes of sampling an inspection lot shall consist of all material of the same type, kind, style, and size submitted for delivery at one time.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. Examination of the end item shall be in accordance with the classification of defects set forth below. The sample size shall be determined by inspection level I, and the acceptable quality level (AQL) expressed in defects per 100 units shall be 1.5. The sample unit for this examination shall be one fitting. Defects shall be classified as shown in table I.

TABLE I. Classification of defects

Examine	Defects
Appearance, construction and workmanship	Not the specified type, kind, size or style of fittings. Unsoundness. Cold shuts. Finish not as specified. Blowholes. Threads not as specified. Nonconformity with dimensions and tolerances.
Marking	Omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

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4.4 Sampling for tests.

4.4.1 Sampling for air pressure test. A random sample of fittings for the air pressure test specified in 4.5.1 or the alternate test specified in 4.5.2, as desired shall be taken from each lot in accordance with inspection level S-3 of MIL-STD-105. The AQL shall be 4.0 percent defective units.

4.5 Test methods.

4.5.1 Air pressure test. Each of the sample fittings selected in accordance with 4.4.1 shall be subjected to an air-pressure test of not less than 60 p.s.i. while the samples are under water or another suitable medium. Fittings shall not leak air at any part of the surface.

4.5.2 Hydrostatic-pressure test. When specified (see 6.2), sample fittings selected in accordance with 4.4.1 shall be subjected to the following hydrostatic test pressure for one minute:

For 125 pound fittings - 250 p.s.i.
For 150 pound fittings - 300 p.s.i.

Under the hydrostatic pressure test, the fittings shall not leak or sweat at any part of the surface.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking of the fittings shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Fittings shall be packaged in accordance with the applicable level A requirements of MIL-V-3.

5.1.2 Level C. The union shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

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5.1.2.1 For Navy procurement subject items of like size and description shall be preserved and packaged in commercial quantities according to the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Fittings shall be packed in accordance with the applicable level A requirements of MIL-V-3.

5.2.2 Level B. Fittings shall be packed in accordance with the applicable level B requirements of MIL-V-3.

5.2.3 Level C. Fittings shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable (see 6.2).

6. NOTES

6.1 Intended use. Bushings, locknuts, and plugs are intended for use with threaded piping at working pressures between 125 and 150 pounds.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Type, kind and style of fittings required (see 1.2.1 and 3.3).

c. Size, head style and dimensions of plugs required (see 3.4).

d. Thickness of finish and type of finish required (see 3.5).

e. When left-hand threads are required (see 3.6).

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f. Whether outside head bushing is required and the dimension of the oversize thread (see 3.6.1).

g. When hydrostatic testing shall be performed (see 4.5.2).

h. Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

i. Marking required (see 5.3).

6.3 All bushings, and plugs made of brass or bronze are covered by WW-P-460. These fittings were formerly type I in the "A" revision.

MILITARY INTEREST:

Custodians:
Army - ME
Navy - YD
Air Force - 82

Review activities:
Army - ME
Air Force - 82

User activities:
Army - WC, MU, CE
Navy - MC

Preparing activity:
Navy - YD

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain copies and other documents referenced herein. Price 15 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
WW-P-471B; Pipe-Fittings, Bushings, Locknuts and Plugs; Iron Steel, and Aluminum (Threaded) k25-150 Pounds

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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